# Quiz 4: Qualitative Analysis III — Mixtures & Equations

*Review of Session 4. ~15 minutes. A calculator helps for Section C.*

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## Section A: Mixtures

1. A sample fizzes a little with acid and leaves white residue that won't dissolve and is iodine-negative. One powder or a mixture? Of what?
2. You want to separate a salt + cornstarch mixture. Describe the steps.
3. In that separation, which component is in the **filtrate** and which is the **residue**?

## Section B: Balancing

Balance by adjusting coefficients only:

1. `__ H₂ + __ Cl₂ → __ HCl`
2. `__ Al + __ O₂ → __ Al₂O₃`
3. `__ CH₄ + __ O₂ → __ CO₂ + __ H₂O`
4. `__ NaHCO₃ + __ HCl → __ NaCl + __ H₂O + __ CO₂`
5. Why can't you balance an equation by changing a subscript?

## Section C: Moles

1. What is the molar mass of $CO_2$? (C = 12, O = 16)
2. What is the molar mass of $NaHCO_3$? (Na = 23, H = 1, C = 12, O = 16)
3. What is the formula: moles = ______ ÷ ______?
4. In a reaction, how do you find the limiting reactant?
5. You have 0.30 mol of reactant A (coefficient 2) and 0.20 mol of reactant B (coefficient 1). Which is limiting?

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## Answer Key

### Section A

1. A mixture — a carbonate (like baking soda) plus an insoluble, non-starch powder (sand or gypsum).
2. Add water and stir (salt dissolves, cornstarch doesn't); filter; evaporate the filtrate to recover the salt; dry the residue on the filter to recover the cornstarch; test each separately.
3. Salt is in the filtrate (dissolved); cornstarch is the residue (on the filter).

### Section B

1. 1, 1, 2 2. 4, 3, 2 3. 1, 2, 1, 2 4. 1, 1, 1, 1, 1
2. Changing a subscript changes what the substance *is* ($H_2O$ vs $H_2O_2$). Only coefficients (the numbers in front) may change.

### Section C

1. 12 + 16 + 16 = 44 g/mol.
2. 23 + 1 + 12 + 48 = 84 g/mol.
3. moles = mass ÷ molar mass.
4. Divide each reactant's moles by its coefficient; the smallest result is the limiting reactant.
5. A: 0.30 ÷ 2 = 0.15. B: 0.20 ÷ 1 = 0.20. 0.15 is smaller → **A is limiting**.